Customer Churn Project

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Problem Description

- US E-Commerce sales stood at $1.19 trillion in 2024
- Major online retail (E commerce) company wants to know the customers who are going to churn, so it can accordingly approach customers to offer some promos.
- Identify factors driving customer churn such as Tenure, CashbackAmount, WarehouseToHome distance, etc
- Goal: Predict likelihood a customer will churn next month

Data Description

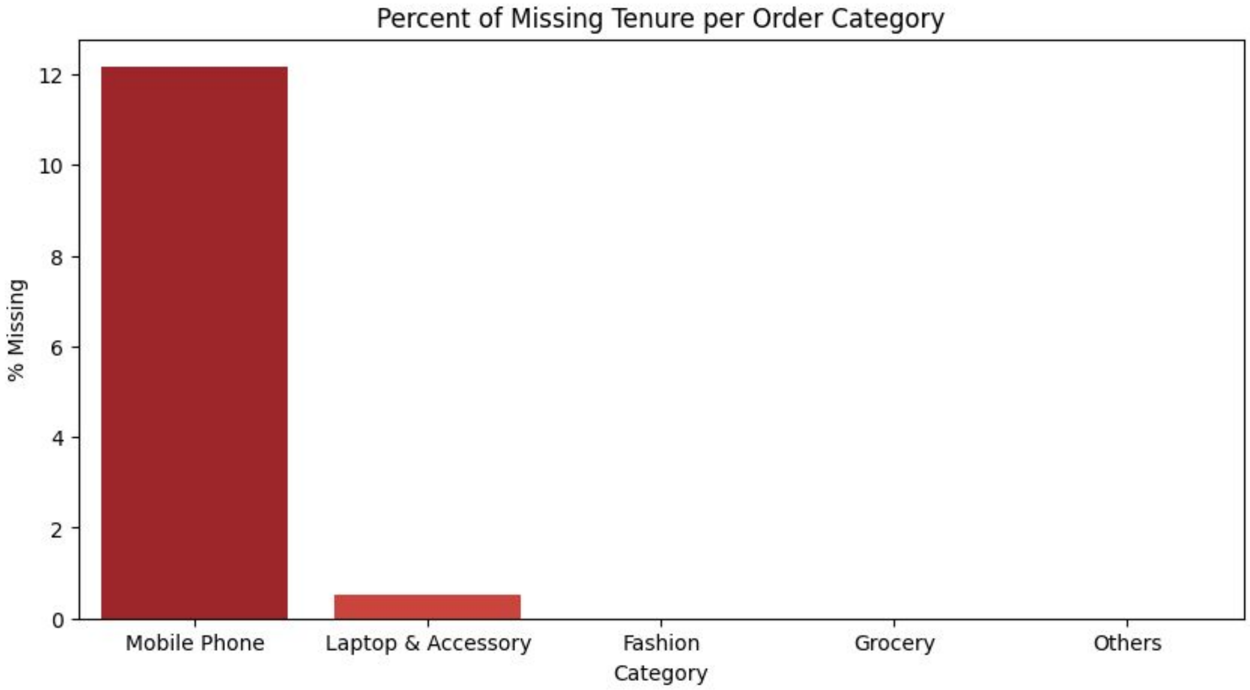
Variable	Discription
CustomerID	Unique customer ID
Churn	Churn Flag
Tenure	Tenure of customer in organization
PreferredLoginDevice	Preferred login device of customer
CityTier	City tier
WarehouseToHome	Distance in between warehouse to home of customer
PreferredPaymentMode	Preferred payment method of customer
Gender	Gender of customer
HourSpendOnApp	Number of hours spend on mobile application or website
NumberOfDeviceRegistered	Total number of deceives is registered on particular customer
PreferedOrderCat	Preferred order category of customer in last month
SatisfactionScore	Satisfactory score of customer on service
MaritalStatus	Marital status of customer
NumberOfAddress	Total number of added added on particular customer
Complain	Any complaint has been raised in last month
OrderAmountHikeFromlastYear	Percentage increases in order from last year
CouponUsed	Total number of coupon has been used in last month
OrderCount	Total number of orders has been places in last month
DaySinceLastOrder	Day Since last order by customer
CashbackAmount	Average cashback in last month

Kaggle dataset that provides a comprehensive view of customer behavior and churn in the e-commerce industry. It includes detailed information on demographics, account tenure, device and payment preferences, engagement metrics, satisfaction levels, complaints, and recent purchasing activity that are critical for analyzing retention and predicting churn.

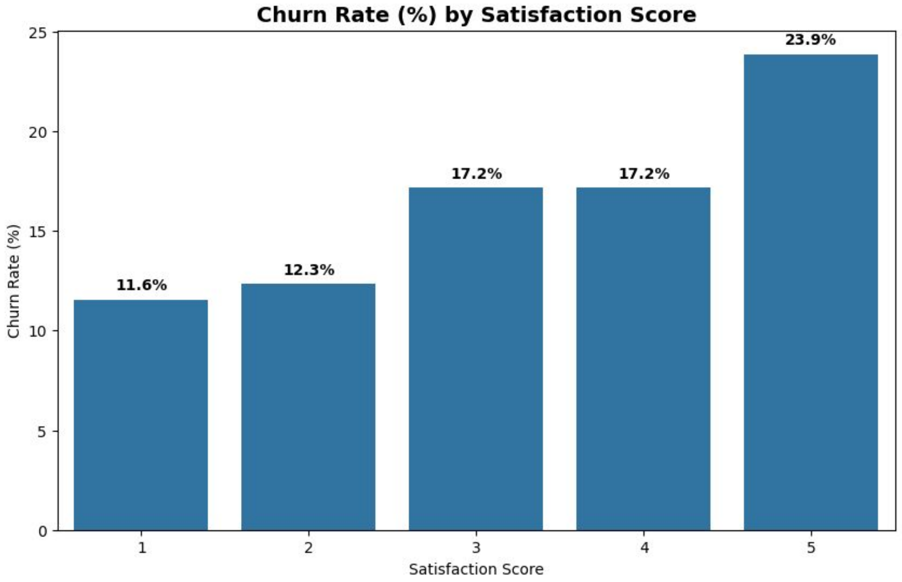
Exploratory Data Analysis (EDA)

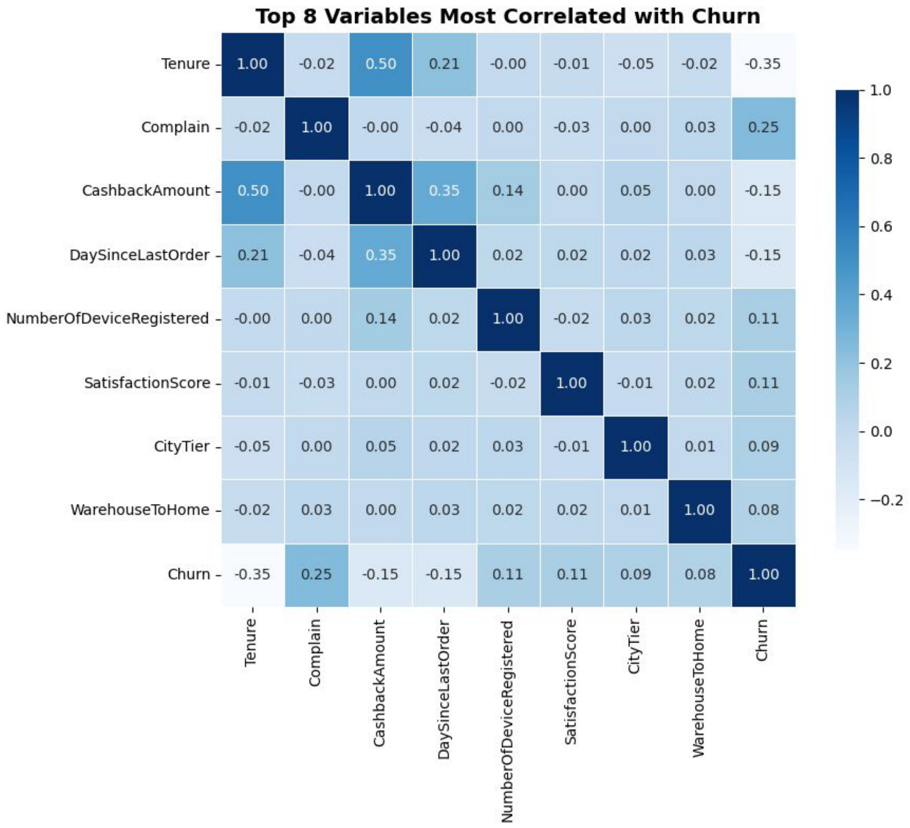
- 5630 observations in total
- High class imbalance in Churn variable (83-17)
- Prioritized keeping as many observations as possible
  - Small dataset size increases value of each observation
  - Impute median values instead of dropping
- Only 13 lost observations (outliers)
  - Final df_cleaned: 5617 observations

EDA Visualizations



Pre-analysis Visualizations





Best Model

Random Forest

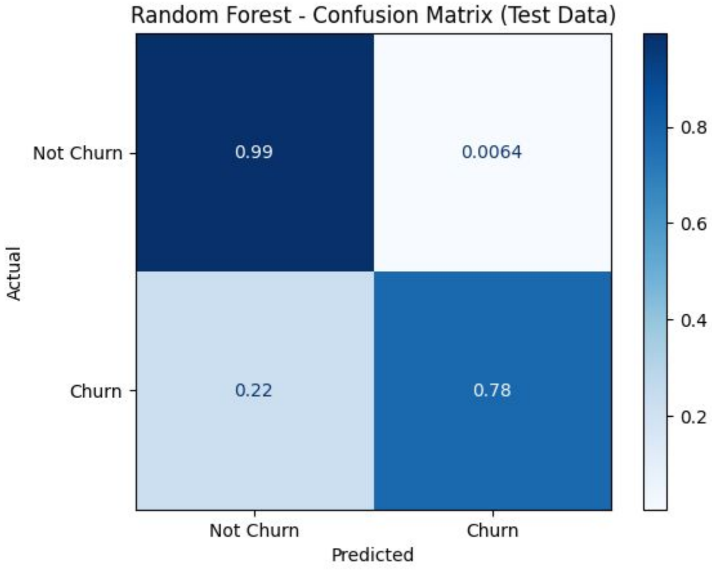
- Best balance between performance and predicting customers who churn
- Best f1-score for customers who churn

What Metrics Matter Most?

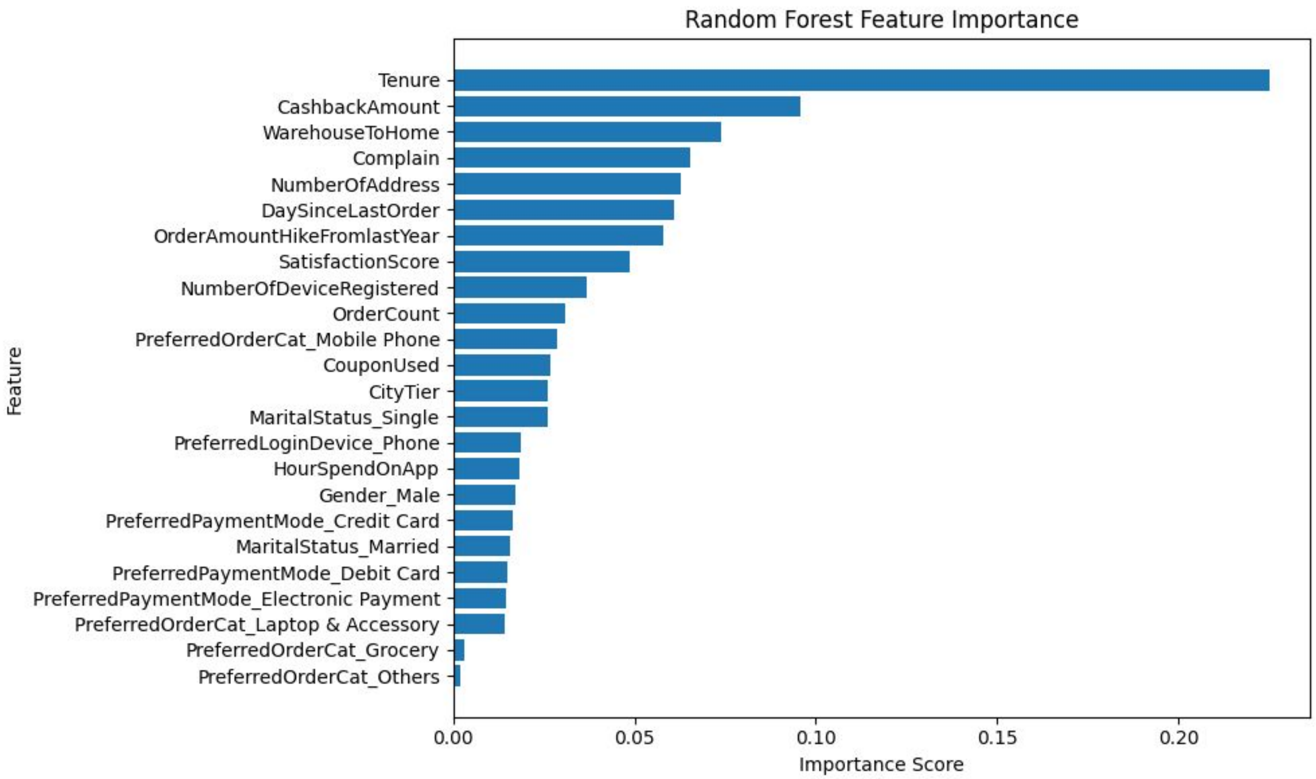
- Recall: Of all customers who actually churned, how many did the model successfully catch?
- Precision: Of all the customers the model predicted as churners, how many actually churn?

Important Metrics

- Train Accuracy: 99.9%
- Test Accuracy: 95.7%
- Non-Churn Recall: 99%
- Churn Recall: 78%
- Non-Churn Precision: 96%
- Churn Precision: 96%



Top Features



Most important feature:

- Tenure

This makes sense intuitively; the longer a customer stays with a company, the less likely they are to leave in the future

Other Important features:

- CashBackAmount
- WarehouseToHome
- Complain

What Actions Should Firms Take? How is the Model Beneficial?

- CashbackAmount is a highly important feature negatively correlated with Churn
    - Maximizing cashback offers may retain customers, lowering Churn likelihood
- WarehouseToHome is also quite important
    - Close warehouse → Fast delivery → Happy customers
- DaySinceLastOrder
    - Frequent orders mean lower churn rate
    - Tempt customers with deals, notifications, etc.

Unimportant Features

- CustomerID - Removed before split
- Order category is unimportant unless the category is Mobile Phone
- PreferredPaymentMode categories are second least important

If Had More Time

- Dive deeper into Neural Network performance
  - Is it really great or too good to be true?
- Explore geographical context more
  - How does WarehouseToHome distance combined with City Tier influence Churn?
  - Potential Folium integration

Thank You!